

LUC MCCUTCHEON

SENIOR RESEARCH SCIENTIST

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PROFILE

Senior Research Scientist specialising in Reinforcement Learning, world models and distributed post-training for embodied robotics. Experience deploying full-stack AI solutions and leading technical teams. Expert in coding algorithms from scratch in PyTorch/JAX, with published research (ICRA, NeurIPS, IROS) focusing on real-world embodied AI challenges.

SKILLS

Frameworks: JAX, PyTorch, Gymnasium, vLLM, Raylib
Code: Python, C++, Rust, JavaScript, Lua, Bash
Methods: World models, PPO/GRPO, SAC, LoRA, VLA, Sim2Real
Compute/Infrastructure: Unitree G1, SLURM, GCP, AWS, Docker
Languages: English (Native), French (C1)

SELECTED TALKS

<i>Guest Speaker & Panel Discussion (Agents Workshop)</i>	Google
<i>Model-based Reinforcement Learning</i>	UoS School of Engineering
<i>JAX vs PyTorch</i>	UoS School of Engineering
<i>Model Optimisation & Compilers</i>	Cambridge Consultants
<i>Safe Reinforcement Learning</i>	Cambridge Consultants

EXPERIENCE

- Research Scientist (Fixed-Term)** Cambridge Consultants (Oct 25 - Apr 26)
- Led the Reinforcement Learning workstream for the Unitree G1 humanoid, delivering robust policies for dynamic environments.
 - Engineered an RL fine-tuning pipeline for VLA models (GR00T 1.6) using PPO/GRPO within custom Isaac Sim environments.
 - Designed MuJoCo and Isaac Sim environments supporting curriculum and continual learning to minimise the sim-to-real gap.
 - Coupled individual foot control with MPC planners to achieve precise, stability-aware foot placement in 3D space.
 - Executed sim-to-real transfer via ONNX to deploy blind stair traversal locomotion on physical hardware.
 - Trained cooperative multi-agent RL policies for a safety-critical defence application, using Kubernetes for distributed training.
- Lead Research Scientist (Part-time)** Agile Loop (Oct 23 - Apr 25)
- Led a team of 10 researchers developing autonomous AI agents, driving technical recruitment and presenting to VC firms.
 - Architected agentic pipelines from scratch and steered the internal research strategy, ranging from fundamental deep learning architecture to RL and VLM LoRA fine-tuning.
 - Presented our full-stack approach to the Google Cloud team during their workshop for AI agents
- Research Scientist (Part-time)** Agile Loop (Jun 23 - Oct 23)
- Pioneered RL environments for software and web apps, using computer vision for icon recognition, and creating systems for distributed multi-task training.
 - Developed software and infrastructure allowing hybrid edge/cloud model routing based on task complexity for Lenovo PoC
- Research Assistant (Volunteer)** Connected & Autonomous Vehicle Lab (Jun 22 - Aug 22)
- Improved exploration using Noisy Neural Networks and tackled partial observability using LSTMs and Deep Q-Learning.
- Software Engineer Intern** QinetiQ (Jul 2018- Sep 2018 / Jul 2019- Sep 2019)
- Engineered tactical voice cryptography (C++/Python) and automated red team attacks across two placements, bridging offensive and defensive security operations.

EDUCATION

- PhD Reinforcement Learning (sponsored by Veolia Nuclear Solutions)**
University of Surrey (2021 - present)
- Designed and implemented time-delay mitigating controller for Veolia Nuclear Solutions' local-remote manipulator (DEXTER).
 - Learned a probabilistic World Model to "undelay" RL environments to provide adaptive PD gains in highly stochastic conditions.
 - Improved stability analysis through learning a Lyapunov function, using RL to generate counter-examples improving robustness.
 - Co-designed and lectured "Intelligent Vehicle Design" university course, teaching planning methods for robotic control
 - Implemented various papers and algorithms from scratch in JAX, such as PCGrad, PPO, ReDo, CBP, VAE, ResNet
 - Implemented numerous custom RL environments, creating complex wrappers for random time-delayed environments and local-remote system parallelism.
- BSc Computer Science (Hons)**
University of Surrey (2018-2021)
- Recipient of the prestigious CyberFirst Scholarship, sponsored by QinetiQ.
 - 1st Class Degree, key modules: Artificial Intelligence (92%), Data Structures & Algorithms (86%), Computational Intelligence (83%)
 - Dissertation in time-series financial forecasting using an LSTM Neural Network with various financial indicators
 - Key takeaways: CUDA programming, evolutionary algorithms and fundamental mathematics.

SELECTED PUBLICATIONS

FIRST AUTHOR

"Calibrated Partial Resets: Preventing Policy Collapse in Continual Reinforcement Learning" — Under Review

- Implemented numerous complex algorithms such as Proximal Policy Optimisation (PPO), Continual Backpropagation (CBP), Recycling Dormant Neurons (ReDo) as well as my own custom optimiser (CPR). I identified parameter resets as a source of instability causing policy collapse in continual reinforcement learning, and improved training stability through soft parameter transformations.

"Neural Lyapunov Function Approximation with self-supervised Reinforcement Learning" — ICRA 2025

- Efficient implementation of probabilistic world model for stable RL under stochastic conditions. This work learns a Soft Actor-Critic (SAC) control policy, world model and Lyapunov function simultaneously, exploiting the geometric properties of the Lyapunov function to analyse system stability.

"Adaptive PD Control using Deep RL for Local-Remote Teleoperation with Stochastic Time Delays" — IROS 2023

- Proposed PMDC, a model-based RL framework utilising a learned world model ensemble to mitigate stochastic delays up to ~300ms.

CO-AUTHOR

"Meta-World+: An Improved, Standardized, RL Benchmark" — NeurIPS 2025 & ICML 2025 Workshop (Spotlight)

- Implemented optimiser algorithms such as PCGrad and GradNorm for multi-task learning

"Exploring In-Context Ensemble with Video-Language Models for Low-Level Workflow Understanding" — NeurIPS 2024 Workshop

- Project definition and implementation of Gemini models into novel in-context ensemble learning framework

"Prediction based decision making for autonomous highway driving" — ITSC 2022

- Implemented baseline "Noisy Neural Networks" and part of the Rainbow algorithm

Reviewer: ICLR 2026, NeurIPS 2025, TNNLS 2025, IROS 2025

All implementations are created from scratch using PyTorch or JAX/Flax trained on a SLURM cluster/ Google Cloud, each demonstrating state-of-the-art performance against baselines on challenging benchmarks.

SIDE PROJECTS

GeoSnap Co-Founder & CTO

An award-winning company which productised AI for information retrieval and navigation, i.e. in large buildings, institutions and universities, where GPS is limited.

- Raised pre-seed funding rounds
- Full-stack serverless web application hosted on Google Cloud
- Web crawler in Rust for data-mining, RAG/LLM pipeline
- Few-shot image classification using CLIP

Freelance

- Tutoring: GCSE to Postgraduate students in Reinforcement Learning, Computer Science & Mathematics
- Industry: Feasibility study on AI communication systems for Cyclopic, delivering a technical report and prototype proposal.

Personal Projects

- Autoreward: LLM RL environment for robotics reward function design
- Facial Recognition Intruder Alert System on Raspberry Pi
- Imitation learning and sensor configuration on autonomous RC cars
- Two-Player Fighting game
- Software automation e.g., using Twilio, PyAutoGUI, OpenCV, etc.
- Capture the flag security competitions and treasurer of information security society
- Web development: Personal website, LLM comparison site, CAV-Lab site and many more

AWARDS AND COURSES

- 3rd at Complex Worlds - (General Reasoning) Hackathon
- Honourable mention - (xAI) Grokathon 2026
- Bronze - Mathematics Olympiad
- Bronze - British Informatics Olympiad
- Grace Hopper Award - Computer Science
- Foundership Award - Student Enterprise
- NVIDIA Fundamentals of Deep Learning
- Coursera Reinforcement Learning Specialisation

INTERESTS

- Neovim and Linux programming
- Reinforcement Learning reading group
- EleutherAI community member
- Life drawing art classes
- London city GO Club (rated 1 Dan)
- Songwriter/guitarist in a local Jazz band
- Sports: rock climbing, running, calisthenics
- Reading: philosophy, psychology, science fiction